

PLASMA SPRAY PROCESSORS

QF/QMS/02/021

Rev. 00

Date : 01/03/2019

Works Instruction



Operations/Work/Job Activity covered by this assessment: Screw

Required Personal Protective Equipment (PPE) :



Safety glasses must be worn at times.



Long and loose hair must be contained.



Appropriate footwear with substantial uppers must be worn.



Close fitting/protective clothing to cover arms and legs must be worn.



Respiratory protection devices required.



Oil free leather gloves and spats must be worn.



Hearing protection must be used when using this machine.

Loading & Unloading Check:

1. Ensure no slip/trip is present in work pass and walk ways.
2. Ensure that work area is clean and clear of grease, oil and any flammable materials.
3. Refer Work Instructions PSP/WI/14 while handling of Screw.

Operations Procedure:

- 1) Refer Pre-Inspection Work Instructions PSP/WI/03 to pre-inspect the Screw.
- 2) If damaged is found on the coating surface, inform customer by email with photograph and get their approval.
- 3) Mask with High Temperature Adhesive Tape/Metal Shim/Insulation tape/Masking Putty to protect unwanted area from blasting and HVOF coating.
- 4) Ensure the Compressed Air is clean & free from water residues.
- 5) Refer Final Blasting Work Instructions PSP/WI/04 for preparing surface. The surface preparation will be as per Sa3 specs.
- 6) Use virgin 24mesh Al₂O₃ grit for blasting. Maintain Air pressure to blast at 3.5bar with 6mm bore nozzle and keeping 200 ±50mm standoff distances.
- 7) Maintain 50 to 75 microns surface profile during blasting and check by using Testex Press-O-Film as per the protocol.
- 8) Load the Screw in the acoustic room and clamp it thoroughly on the chuck with proper packing to prevent clamping marks on the Screw Shaft.
- 9) Start Exhaust Fans before starting the HP/HVOF Control Unit.
- 10) Open valve of Nitrogen and Oxygen Cylinders and set the pressure according to the parameter mentioned in the Technical Bulletin.
- 11) Start the JP8000 control unit with your Login ID. Check the Oxygen, Fuel, Water & Combustion Flow & pressure in the manual mode screen.
- 12) Set the parameter as given by Extreme Coatings for Xc1000, Xc 1000Ni, Xc 1000HD and Xc 4000.

i) Barrel

6"

ii) Oxygen Pressure

240+₅₀PSI

Flow

2000SCFH±50

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iii) Nitrogen Pressure 55PSI Flow 23SCFH±2 iv) Fuel Pressure 119PSI Flow 6GPH±.25 v) Spray Rate 75 to 80gms/min vi) Spray Distance 350±25mm 13) Pre-heat the powder in the powder feeder hopper and maintain the temperature with heating blanket. Check with pyrometer the temperature of the heating blanket and ensure it is at 60°C to 65°C. 14) Select the powder recipe from the recipe menu. 15) Measure the powder flow rate (75 to 80 grams/min) as per parameter, and set the Powder feeder RPM. 16) Set the job speed and Robot programming speed, as given in chart or instructions will be given by Supervisor. 17) Closed the Acoustic Door and Wait for signal before igniting the JP 5000 Torch. Go to Development screen and press the gun ignition. Wait for the gun to ramp up till you get the desire combustion pressure as per the parameter. 18) Pre-heat the skin temperature of the job up to 50°C to 60°C, check with pyrometer (For smaller size screw). 19) Start Powder feeder and wait till it stabilize and spray the job layer by layers to get a uniform thickness all over.					
Shutdown : 1. Shut the Powder Feeder and wait for the feed lines to clear. 2. Shut down the ignition flame and bring the robot to its home position. 3. Measure the dimension as given in the job card and allow it to cool completely before removing from the Spray Room. 4. Refer to Final Inspection Work Instruction PSP/WI/10 and get the approval of Supervisor.					
Assessor :	Prasanna Wadkar	Signature:		Date:	5.04.19
Approved:	Dhimant Doshi	Signature:		Date:	5.04.19

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